Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 10)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows some techniques used to locate mineral deposits.

0 1 . 1 Complete Table 1 by adding the name of the technique or a description of how it works.

[4 marks]

Table 1

Technique	How it works
Gravimetry	
	Detects rocks that are more magnetic, such as magnetite iron ore or ores of tungsten and cobalt
Seismic survey	
	Measures how difficult it is for electricity to pass through rocks; sedimentary rocks with high water content have lower values than igneous rocks

0 1 . 2 Explain why trial drilling is usually carried out only after remote sensing surveys.

[3 marks]

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0 1 . 3 Explain how infrared spectroscopy from satellites can help identify mineral deposits.

[2 marks]

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0 1 . 4 Give **one** advantage of portable field analysis equipment such as X-ray fluorescence.

[1 mark]

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0 2 . 1 Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops ☐

B Steam washing ☐

C Inflatable booms ☐

D Polymerising agents ☐

0 2 . 2 Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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0 2 . 3 Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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0 3 Table 1 shows information about some greenhouse gases.

Table 1

Gas	Main anthropogenic sources	Approximate residence time	Global warming potential (per molecule, relative to CO_2 over 100 years)
Carbon dioxide	Fossil fuel combustion, deforestation, cement	100+ years (variable)	1
Methane	Rice paddies, cattle, landfill, gas leaks, coal mines	About 12 years	28
Nitrous oxide (NOx)	Fertilisers, combustion	About 120 years	265
CFCs	Refrigerants, aerosols (now banned)	50-100+ years	5 000-10 000
Tropospheric ozone	Secondary pollutant from NOx and hydrocarbons	Days-weeks	-

0 3 . 1 Explain the natural greenhouse effect.

[3 marks]

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0 3 . 2 Use Table 1 to explain why carbon dioxide is considered the most important anthropogenic greenhouse gas even though methane and CFCs absorb far more infrared per molecule.

[3 marks]

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0 3 . 3 Explain why reducing methane emissions would produce a faster reduction in warming than reducing CO_2 emissions by the same mass.

[2 marks]

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0 3 . 4 Give **two** methods of reducing methane emissions from agriculture or waste.

[2 marks]

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0 4 . 1 Explain why abstraction from an aquifer containing 'fossil' (palaeo) water, such as the High Plains / Ogallala Aquifer in the USA, cannot be sustainable.

[3 marks]

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0 4 . 2 Describe how the GRACE satellites are used to monitor aquifer depletion and give **one** advantage over monitoring boreholes.

[3 marks]

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0 4 . 3 Describe **three** measures being introduced on the North China Plain to reduce aquifer depletion.

[3 marks]

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0 4 . 4 Suggest why large-scale afforestation on the North China Plain may reduce aquifer recharge even though trees reduce soil erosion.

[1 mark]

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0 5 . 1 Explain how each of the following environmental factors affects the rate of degradation of pollutants. Give a named example of a pollutant for each factor.

[6 marks]

Temperature .....

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Light .....

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Oxygen .....

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0 5 . 2 Explain how the presence of adsorbent materials such as clay can reduce the severity of pollution by heavy metals or radioactive caesium-137.

[2 marks]

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0 5 . 3 Explain why a fall in pH increases the harm caused by many heavy metals.

[2 marks]

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0 6 . 1 Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

A A waste is assumed to cause pollution until research shows it is unlikely to do so ☒

B The polluter must pay for any damage caused ☐

C Emissions should be as low as reasonably achievable ☐

D Only pollutants known to be toxic need to be controlled ☐

0 6 . 2 Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

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0 6 . 3 Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

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0 7 The Universal Soil Loss Equation (USLE) is A = R x K x LS x C x P. Table 1 gives factor values for a farm.

Table 1

Factor	Description	Value
R	Rainfall erosivity for the farm	110
K	Soil erodibility: loam	0.67
LS	Slope length and gradient: 120 m, 6%	1.35
C (crop type x tillage)	Beans (0.50) with spring ploughing (0.90); or wheat (0.35) with mulch tillage (0.60)	see question
P	Ploughing up and down slope 1.0; contour ploughing 0.50; strip cropping with contour ploughing 0.25	see question

Table 2

Soil erosion class	Rate of soil loss / t ha^{-1} yr^{-1}
Low	under 5
Moderate	5-10
High	10-20
Very high	20-30
Severe	over 30

0 7 . 1 Calculate the annual soil loss for beans with spring ploughing, ploughed up and down the slope. Give your answer to one decimal place.

[2 marks]

Answer _____ t/ha/yr

0 7 . 2 Calculate the annual soil loss for wheat with mulch tillage and strip cropping with contour ploughing.

[2 marks]

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Answer ____ t/ha/yr

0 7 . 3 The natural rate of soil formation on the farm is 1 t ha^{-1} yr^{-1}. Use your answers and Table 2 to evaluate the sustainability of the two farming systems.

[3 marks]

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0 7 . 4 Explain why the USLE gives only an estimate of soil loss.

[2 marks]

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0 7 . 5 Suggest which single factor in the USLE a farmer can change most easily.

[1 mark]

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0 8 Table 1 shows the mean surface temperature of the Earth with and without an atmosphere containing greenhouse gases.

Table 1

Condition	Mean surface temperature / °C
No greenhouse effect	-18
Natural greenhouse effect	15

0 8 . 1 Use Table 1 to calculate the warming produced by the natural greenhouse effect.

[1 mark]

Answer °C

0 8 . 2 Explain how the natural greenhouse effect warms the Earth's surface.

[4 marks]

0 8 . 3 Explain what is meant by a **greenhouse gas**.

[1 mark]

0 8 . 4 The wavelengths of radiation arriving at the Earth are different from those leaving. Describe this difference and explain why it matters to the energy budget.

[2 marks]

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0 8 . 5 Explain why a change in the movement of energy through the atmosphere could affect the distribution of species.

[2 marks]

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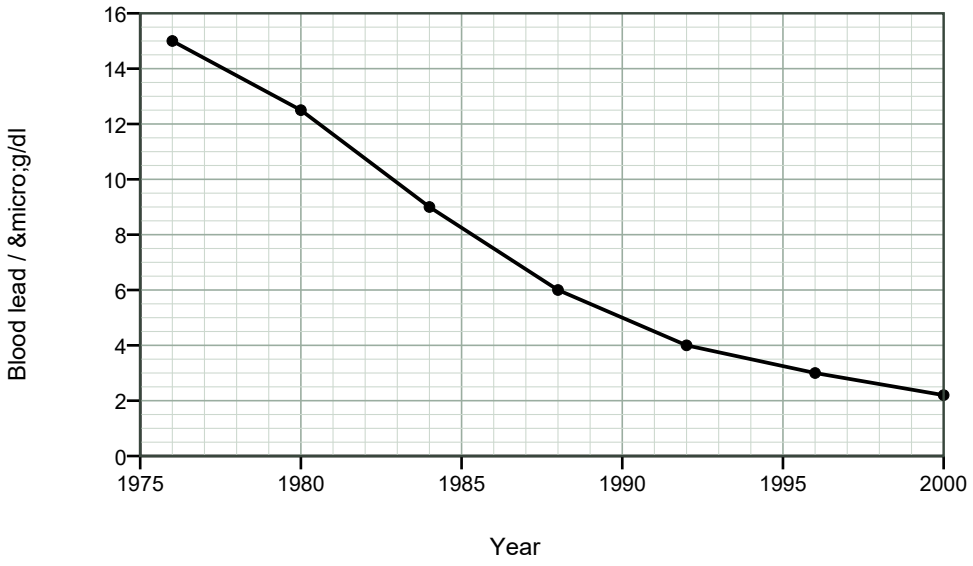
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0 9 Figure 1 shows the mean concentration of lead in the blood of children in a large city between 1976 and 2000, and the year in which lead additives in petrol began to be phased out.

Figure 1



0 9 . 1 Calculate the percentage decrease in mean blood lead between 1976 and 2000.

[2 marks]

Answer [ ] %

0 9 . 2 Explain why tetraethyl lead was added to petrol and why this caused a public health threat.

[3 marks]

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0 9 . 3 Suggest **two** reasons, other than petrol additives, why blood lead concentrations continued to fall after 1990.

[2 marks]

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0 9 . 4 Explain how adding small amounts of phosphoric acid to a water supply reduces lead in drinking water.

[2 marks]

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0 9 . 5 Explain why discarded lead fishing weights and shotgun pellets can poison waterfowl.

[1 mark]

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1 0 A student investigated whether the number of earthworms per m^2 differs between an organic field and a conventionally farmed field. Table 1 shows the results.

Table 1

Field	Number of samples	Mean earthworms per m^2	Standard deviation
Organic	10	48	14
Conventional	10	31	12

1 0 . 1 Name the statistical test the student should use and explain why it is appropriate.

[2 marks]

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1 0 . 2 The test gave p = 0.01. State the conclusion the student should draw.

[2 marks]

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1 0 . 3 Explain why the student cannot conclude that organic farming **caused** the higher earthworm numbers.

[2 marks]

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1 0 . 4 Evaluate the decisions that must be made when planning a fieldwork investigation so that valid conclusions can be drawn.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

OR

1 1 . 2 Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS